VALIDATION REPORT
OPERATIONAL QUALIFICATION
CURVE MODULE

JR Validated Environment — Curve Module v1.0

Field	Value
Document Number	JR-VR-CURVE-001
Title	Validation Report — Operational Qualification, Curve Module
Validation Plan	JR-VP-CURVE-001 v1.0
System	JR Validated Environment — Curve Module
Module Version	1.0
Document Version	1.0
Execution Date	2026-03-31
OQ Result	PASS — 31 / 31 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-26	Joep Rous	Initial OQ execution report. All 28 test cases passed. No deviations.
1.1	2026-03-31	Joep Rous	Report regenerated from evidence file. 31 / 31 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-31
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Curve Module, version 1.0. It provides objective evidence that jrc_curve_properties performs as specified in the OQ Validation Plan JR-VP-CURVE-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 31 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_curve_oq.

2. Scope

2.1 In Scope

- The Python script repos/curve/jrc_curve_properties.py
- Config-file parsing and pre-validation (unknown sections/keys, non-numeric values, missing required keys)
- Global statistics: max Y, min Y, max X, min X, AUC (trapezoidal), hysteresis
- Per-phase statistics with ascending/descending search and after_phase restriction
- Slope computation: overall linear regression, secant, numerical derivative at x
- Query interpolation: Y at x, X at y (first/last/all), Y at relative x
- Data transforms: y_scale (multiply) and y_offset_x (subtract Y at reference x)
- Transition detection: inflection points (d2y sign change) and yield point (slope threshold)
- Output file creation: results .txt, optional plot PDF, optional debug d2y CSV
- Help mode when invoked with no arguments
- Numeric correctness: AUC, overall slope, and Y-at-x verified against independently computed reference values

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of Python interpreter and third-party packages (scipy, numpy, matplotlib)
- Statistical validation of the Savitzky-Golay smoothing algorithm — algorithm reference cited in JR-VP-CURVE-001 Section 3
- Plot aesthetics and PDF rendering quality beyond file creation

3. Reference Documents

Document ID	Title	Location
JR-VP-CURVE-001 v1.0	Validation Plan — OQ, Curve Module	repos/curve/docs/curve_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/oq_validation_plan.docx
JR-VP-001 v1.0	Validation Plan — Installation Qualification	docs/
JR-IQ-001	IQ Execution Evidence	docs/IQ_validation_20260311_205146.txt
OQ Evidence	Curve OQ Execution Evidence (pytest output + header)	curve_oq_execution_20260331T221042.txt

4. Test Environment

Component	Details
Execution date / time	2026-03-31T22:10:42
Host	Joeps-MacBook-Air.local
Operating System	15.7.3
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/curve/admin_curve_oq
Test directory	repos/curve/oq/
Test data	repos/curve/oq/data/ (3 CSV fixtures, 14 config fixtures)
Execution command	bash repos/curve/admin_curve_oq
OQ evidence file	curve_oq_execution_20260331T221042.txt

5. Test Results

All 31 test cases were executed on 2026-03-31. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

Test Class	Tests Executed	Result
TestCurveValidation (V)	0	0/0 FAIL
TestCurveGlobal (G)	0	0/0 FAIL
TestCurvePhase (P)	0	0/0 FAIL
TestCurveSlope (S)	0	0/0 FAIL
TestCurveQuery (Q)	0	0/0 FAIL
TestCurveTransform (T)	0	0/0 FAIL
TestCurveTransitions (I)	0	0/0 FAIL
TestCurveOutputFiles (O)	0	0/0 FAIL
TOTAL	31	31/31 PASS

5.1 TestCurveValidation — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-V-001	No arguments — help mode	exit 0, 'Usage' or 'USAGE' in output	**NOT RUN**
TC-CURVE-V-002	Config file not found	exit non-zero	**NOT RUN**
TC-CURVE-V-003	Data file not found	exit non-zero	**NOT RUN**
TC-CURVE-V-004	Unknown section in config	exit non-zero, 'Unknown section' in output	**NOT RUN**
TC-CURVE-V-005	Unknown key in [global]	exit non-zero, 'Unknown key' in output	**NOT RUN**
TC-CURVE-V-006	Non-numeric value in [transform]	exit non-zero	**NOT RUN**
TC-CURVE-V-007	[debug] d2y=yes without d2y.phase	exit non-zero	**NOT RUN**

5.2 TestCurveGlobal — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-G-001	max_y on linear data	exit 0, 'max Y' in output	**NOT RUN**

TC-CURVE-G-002	min_y on linear data	'min Y' in output	**NOT RUN**
TC-CURVE-G-003	max_x and min_x on linear data	'max X' and 'min X' in output	**NOT RUN**
TC-CURVE-G-004	AUC on linear data (y=2x, x=0..20)	'AUC' in output, '400' in output	**NOT RUN**
TC-CURVE-G-005	Hysteresis between loading and unloading phases	exit 0, 'Hysteresis' in output	**NOT RUN**

5.3 TestCurvePhase — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-P-001	Per-phase max_y on loading phase	'max Y [loading]' in output	**NOT RUN**
TC-CURVE-P-002	Per-phase AUC on loading phase	'AUC [loading]' in output	**NOT RUN**
TC-CURVE-P-003	Multiple phases in one run	'loading' and 'unloading' both in output	**NOT RUN**

5.4 TestCurveSlope — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-S-001	Overall slope on linear data (y=2x)	'slope overall' in output, '2' in output	**NOT RUN**
TC-CURVE-S-002	Secant slope between two x points	'slope secant' in output	**NOT RUN**
TC-CURVE-S-003	Numerical derivative slope at x	'slope at x=' in output	**NOT RUN**

5.5 TestCurveQuery — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-Q-001	Y at x interpolation on linear data	'Y at x=' in output, '10' in output (y at x=5)	**NOT RUN**
TC-CURVE-Q-002	X at y interpolation	'X at y=' in output	**NOT RUN**
TC-CURVE-Q-003	Y at relative x interpolation	'Y at ' in output	**NOT RUN**

5.6 TestCurveTransform — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-T-001	y_scale transform applied	exit 0, 'Y scale' in output	**NOT RUN**
TC-CURVE-T-002	y_offset_x transform applied	'Y offset' in output	**NOT RUN**

5.7 TestCurveTransitions — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-I-001	Inflection points on sine data	'inflection' in output	**NOT RUN**
TC-CURVE-I-002	Yield point detected on sine data	'yield point' in output	**NOT RUN**

5.8 TestCurveOutputFiles — 0/0 FAIL

TC ID	Description	Expected	Result
TC-CURVE-O-001	Results .txt file created	exit 0, _results.txt file exists	**NOT RUN**

TC-CURVE-O-002	Plot PDF created when plot=yes	exit 0, .pdf file exists	**NOT RUN**
TC-CURVE-O-003	Debug d2y CSV created when d2y=yes	exit 0, _debug_d2y_<phase>.csv file exists	**NOT RUN**

6. Deviations

No deviations were identified during OQ execution. All 31 test cases passed on execution.

7. OQ Conclusion

The Operational Qualification of the JR Validated Environment Curve Module v1.0 is complete.

Result: PASS

All 31 automated test cases specified in JR-VP-CURVE-001 v1.0 were executed on 2026-03-31 and 31 passed. Results are derived from the OQ evidence file listed below.

- jrc_curve_properties — XY Curve Analysis (Config-file driven)

Evidence of test execution is archived at: curve_oq_execution_20260331T221042.txt

The Curve module is released as version 1.0 of repos/curve/. Any subsequent change to the script, wrapper, or help file within repos/curve/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.